

PANCHADIP BHATTACHARJEE

☎ 8116760429 ✉ panchadip125@gmail.com 🌐 GitHub: <https://github.com/Panchadip-128> 🌐 Website
🌐 <https://www.linkedin.com/in/panchadip-bhattacharya-192388228/> 🌐 Leetcode: <https://leetcode.com/u/Panchadip/>

EDUCATION

- **Bachelor of Technology - Computer Science and Engineering (Artificial Intelligence)** 07/2023 - 07/2027
Manipal Institute of Technology- Bengaluru , Manipal Academy of Higher Education- Manipal SGPA : 8.67/10.0, CGPA : 8.0/10

SKILLS

Languages: Python, Java, C++, SQL. **AI/ML :** PyTorch, TensorFlow, Scikit-Learn, HuggingFace Transformers, BERT, XGBoost, NLP, LLM, Fine-tuning **Backend & APIs:** Flask, FastAPI, Django, REST APIs, Node.js **Web Development:** React.js, Next.js, Tailwind CSS, Streamlit **Cloud:** Git, Github, Docker, Kubernetes, Linux, AWS, Azure.

PERSONAL PROJECTS

- **CodeMate AI - Advanced LLM Copilot Extension- VS Code AI-Powered Coding Assistant**
 - Created an AI-powered VS Code extension using Llama 3.1, GPT-3.5-turbo, and Gemini 1.5 to generate code snippets, solve LeetCode/DSA/CP problems, and boost coding efficiency by 40%.
 - Enabled seamless model switching and real-time AI assistance, boosting development speed by 40% and accuracy by 60%.
 - **Tech Stack:** TypeScript, React, CSS, LLM models (Llama 3.1, GPT-3.5-turbo, Gemini 1.5), VS Code, Docker, Kubernetes.
- **Risk-Intelligence-Multi-Dimensional Health Assessment with Machine-Learning**
 - Constructed an AI-driven health risk assessment tool using Streamlit and Scikit-learn, predicting five critical diseases with 85–92% accuracy and enabling 30% faster early detection through intuitive visual analytics.
 - **Tech Stack:** Seaborn, Pandas, RandomForest Classifier, Streamlit, Matplotlib, Numpy
- **Job-Finderr – Scalable Job Discovery platform**
 - Developed a full-stack job portal using Next.js 14 (TypeScript) and Node.js, with modular architecture and optimized APIs; reduced render latency by 40% .
 - Deployed microservice architecture with Next.js (Vercel) and Express (Render), integrated MongoDB Atlas, and improved API response time by ~30% with secure CORS and route protection. – Designed RESTful APIs and optimized backend performance, reducing response latency by ~30%
- **MCP-First-Collaborative Multi-Agent Workflow Platform**
 - An open-source developer copilot platform with MCP orchestration, hybrid RAG (vector + BM25), and streaming chat, enabling multi-agent workflows (planner → coder → reviewer → tester) containerized using Docker and Kubernetes
 - Built LLM powered AI agent workflows for contextual analysis and intelligent decision support , testing and validation for multi-agent workflows to ensure reliability. **Tech Stack:** FastAPI, React, Weaviate, Redis, Tailwind, TypeScript, Docker

EXPERIENCE

- **Project Trainee Intern – Intel Corporation** 05/2025- 08/2025
 - Developing a tag generation pipeline using the Qwen2-1.5B Instruct LLM to automatically classify research papers across 50+ categories, increasing enrichment speed by 70% and enabling large-scale topic discovery deployed using Kubernetes.
 - Developed a modular, end-to-end sentiment analysis system with 6+ microservices (PII anonymization, profanity masking, sentiment classification, topic modeling), deployed via Docker, FastAPI, and Streamlit, improving review processing speed by 65% and enabling real-time feedback analysis across 1000+ entries.
- **Full Stack SDE Intern – Infosys** 11/2024 - 02/2025
 - Implemented a full-stack website enabling 35% faster data-driven decision-making with real-time analytics, visualization, sentiment analysis, and anomaly detection and fused database rendering, role-based authentication, route optimization, and sales trend analysis, improving efficiency by 40%.
 - Boosted user engagement by 50% with real-time notifications and improved security by 60% using email MFA.
Tech used: Flask, Tailwind, React JS, MySQL, Streamlit, AWS, Docker
- **AI & Data Science Intern- Coding Jr** 06/2024 - 08/2024
 - Fine-tuned LLM-based Copilots using Llama 3.1, GPT-3.5 & Gemini 1.5, integrating Groq & Together.ai APIs into a VS Code extension and standalone tool - accelerated dev productivity by 30%.
Tech leveraged: Groq, VS Code, Llama ,GPT engines.
- **SDE Intern -Bluestocks Fintech** 10/2024 - 01/2025
 - Executed key features like IPO details, price bands, stock status, issue size, listing date, listing gain, CMP, current return, enhancing user engagement by 35% and decision-making efficiency by 50%.
Tech employed: Django- REST Framework, PostgreSQL, Bootstrap, React JS, AWS, Docker

CERTIFICATIONS AND CREDENTIALS

- **Microsoft Certified Azure AI 900 Professional-** Scored a total of 942/1000 in the final proctored examination
- **Amazon ML School Mentee** - Selected among top 3000 candidates out of 60,000+ candidates, mentored by Amazon scientists and team on different industry ML projects and applications.
- **Intern- NIT K** - Worked on MERN Stack Development applications as a part of Internship project in collaboration with academic partners of Pantech Eduventures Pvt Ltd and NIT Surathkal STEP(NIT K's Science & Technology

Entrepreneurs Park) professors and student developers

- **Data Science Intern – International Institute of SDG's and Public Policy Research** - Conducted data-driven research on public policy, emerging technologies, semiconductor ecosystems, analyzing datasets and policy trends.

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY:

- Ranked **124 among 50k + candidates** in Open Source Contributions at Girls Script Summer of Code 2024.
- **AICTE Student Scholar**- Received a grant of ₹11,00,000 as a tuition fee waiver for academics based on merit.
- Vice Chair at ACM St. Chapter, **ACM India Member**, *Papers accepted & presented at IEEE CONECCT'25 – flagship conference of IEEE BLR Section , HiPC- SRS , IEEE INDICON- flagship conference of IEEE India Section, COMSNET (Poster Track), INDOSYS*

PUBLICATIONS

HAMP-Net: GPU-Accelerated Hierarchical Adaptive Mixed-Precision Training with Gradient Compression for Image Classification

IEEE 32nd International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW) – (Part of Core A)

- Proposed **HAMP-Net**, a hierarchical neural architecture for efficient deep learning training in distributed and high-performance computing environments.
- Introduced **HAMP-GC**, an adaptive gradient compression mechanism reducing communication overhead by **85–95%** in distributed training pipelines.
- Integrated **mixed-precision training (FP16/FP32)** and hierarchical residual blocks with attention mechanisms to maintain training stability.
- Evaluated on **CIFAR-10 and CIFAR-100**, achieving **78.32% accuracy** with significantly faster training on Tesla T4 GPUs.
- Demonstrates a scalable framework for **efficient large-scale deep learning training and HPC-oriented AI systems**.

Dual-Stream Contrastive Graph Neural Network with Laplacian Positional Encoding for Robust Node Classification

IEEE 32nd International Conference on High Performance Computing, Data and Analytics Workshop (HiPCW) – (Part of Core A)

- Proposed **DSC-GNN**, a dual-stream graph neural network architecture integrating **structural and diffusion-based graph representations**.
- Augmented node features with **Laplacian Positional Encoding (LPE)** to capture graph topology.
- Combined **attention-based stream fusion with contrastive learning (InfoNCE)** to improve representation robustness.
- Achieved **80.9% node classification accuracy on the Cora citation network**.
- Validated embedding quality using **t-SNE visualizations, confusion matrix analysis, and ablation studies**.

QUIBC: A Quantum-Inspired Image Binarization Compressor for Resource-Constrained Edge Devices

IEEE 22nd India Council Conference -INDICON 2025 - Flagship Conference of IEEE India Council

- Proposed **QUIBC**, a quantum-inspired deep learning architecture for efficient image compression on **edge and IoT devices**.
- Utilized **unitary transformations and latent-space binarization** for computationally efficient feature encoding.
- Optimized compression via **adaptive rate-distortion loss with a learnable Lagrange multiplier**.
- Achieved **26.58 dB PSNR and 0.914 MS-SSIM at 0.45 bpp** on the CLIC dataset.
- Demonstrated deployment feasibility on **Coral Edge TPU and NVIDIA Jetson devices**, highlighting potential for real-time edge AI applications.
- Available on IEEE Xplore: <https://ieeexplore.ieee.org/document/11392962>

Exposing Digital Illusions: A Comparative Analysis of CNN Architectures for Deepfake Detection

IEEE 11th International Conference on Electronics, Computing and Communication Technologies (CONECCT)- 2025 – Flagship Conference of IEEE Bangalore Section

- Conducted a comparative study of **VGGFace16, DenseNet-121, and a custom CNN** for deepfake detection.
- Trained models on **140k+ real and synthetic face images** with extensive preprocessing and augmentation pipelines.
- Achieved **>95% classification accuracy and ROC-AUC ≈ 0.99** across evaluated architectures.
- Integrated **Explainable AI techniques (SHAP, LIME)** to interpret model predictions and highlight discriminative facial features.
- Demonstrates effective **AI-based approaches for digital media verification and misinformation detection**.
- Available on IEEE Xplore: <https://ieeexplore.ieee.org/document/11306625>

Multi-Objective Edge Resource Optimization with D3QN-PER in Quantum-Classical IoT Edge Networks

IEEE COMSNETS 2026 - Poster Track (CORE C Conference)

- Proposed **D3QN-PER**, a reinforcement learning framework for adaptive resource scheduling in heterogeneous edge computing systems.
- Combined **Dueling Double Deep Q-Network architecture with Prioritized Experience Replay** to improve learning stability.
- Developed a **custom OpenAI Gym simulation environment** modeling dynamic IoT edge workloads.
- Optimized multiple objectives including **latency, load balancing, node utilization, and overload prevention**.
- Demonstrated significant improvements over heuristic scheduling strategies such as **Round-Robin and Least-Loaded algorithms**.
- Available on IEEE Xplore: <https://ieeexplore.ieee.org/document/11418298/>